ZR3I

Zr3i MVP: Development Plan v2

C# and MongoDB, multi-tenant, with the risk module. Version 2, 15 October 2026. Prepared for Ahmad.

The MVP is a multi-tenant SaaS with two modules, Verification (R0) and Risk (CRI per parcel, FRI per farm), built in about six weeks on one stack: .NET 10 and MongoDB. This is v2 of the plan; v1 ("Zr3i R0: Development Guidelines and Plan") stays as it was. Where v2 is silent, v1 still applies, including its hard rules and domain rules for Verification.

What changed from v1

Topic	v1	v2
Customers	One pilot customer drives scope	Generic engine with five built-in account types as presets: Zr3i admin, Bank, Insurance, Company, Farmer. No per-type code paths
Tenancy	Application-level gate, deferred to customer 2	Tenant id on every document from day 1, enforced in the data layer, proven by a two-tenant test
Risk index	Not now	In the MVP as its own module, always labelled "Specified · not calibrated", method FRI-2026-01
Entities	Parcel and parcel-season	Organisation, Portfolio, Farm, Parcel, ParcelSeason, custom fields, external ids
Scale	About 450 parcels	Bulk import, background jobs and cached scene statistics for very large books (thousands of farms)
Schedule	90 days, a gate per phase	About 6 weeks, three gates
UI	Not specified	Design canvas published: 44 boards, English and Arabic, field-green identity, a view for each account type

The full target design is in the separate document "Zr3i Platform Architecture: Multi-tenant, Generic, Expandable". This plan is the build order for it.

MVP scope

- **In:** organisations, users, roles, the five account types (Zr3i admin, Bank, Insurance, Company, Farmer) with direct and company-managed farmer accounts, sign-up by account type, invitation sign-in, cross-tenant grants for sharing, farms and parcels with bulk import, Verification R0 with evidence and accuracy status, Risk CRI and FRI with methodology page, portfolio and farm views, English and Arabic UI and PDF reports, audit log, consent gate.
- **Later:** single sign-on, billing, public API, calibrated indexes, mobile app, monitoring and alerts, a module marketplace.

Account types

Five built-in account types, each a preset (default roles, navigation, enabled modules, home view, sign-up fields) on the same tenant engine. Core code never branches on the type.

Type	Model	Reads other tenants' data
Zr3i admin	Provider	Only in a logged support session
Bank	B2B lender	Only with a grant
Insurance	B2B insurer	Only with a grant
Company	B2B tenant, or an authority such as AgriCash	Only what it is granted
Farmer	B2B and B2B2C: direct, or managed by a company or authority	No. Owns consent and can withdraw any grant

- A managing company can add and update a farmer's farms and parcels, and read results only where consent has been given. Every read is logged.
- Each grant to a bank, insurer or company has a purpose, a period and a Withdraw action.
- A farmer without an account can still give consent through a one-page link in Arabic or English.

Architecture and solution layout

One .NET solution, one deployable web app plus one worker, one MongoDB database. A modular monolith: modules are projects with a manifest, not services.

```
src/
  Zr3i.Kernel/          tenancy, users, roles, grants, audit, jobs, custom fields, module
registry
  Zr3i.Geo/             NetTopologySuite, ProjNet, GDAL window reads, pixel test, zonal stats
  Zr3i.Modules.Verification/   R0 rules, findings, evidence, accuracy register
  Zr3i.Modules.Risk/           CRI, FRI, bands, indicator results
  Zr3i.Web/             ASP.NET Core API + Razor/Blazor UI, EN/AR, RTL
  Zr3i.Worker/          job runner: import, imagery, indicators, reports
  Zr3i.Reports/         HTML to PDF, CSV, SHA-256 manifest
tests/                  unit, golden datasets, two-tenant isolation, contract tests
```

Rules that save time later

1. Every collection has `tenantId`. Repositories take a `TenantScope` argument and add the filter; no code touches the driver directly. A test with two tenants fails the build if any read crosses.
2. A module registers a manifest: entity types, permissions `module:resource:action`, indicators, jobs, nav items, report templates. The kernel loads manifests, so a new module needs no kernel change.
3. Indicators are immutable results: method version, input hash, value, components, missing components, maturity (`specified`, `calibrated`, `validated`).
4. Append-only collections (audit, consent, verification, evidence, indicator results) use an application role with find and insert only.
5. Bulk work goes through a durable job queue in MongoDB with per-tenant fairness. Nothing heavy runs in a web request.
6. UI built from the design system tokens; every string in a resource file with English and Arabic from the first screen.

Indexes and scale

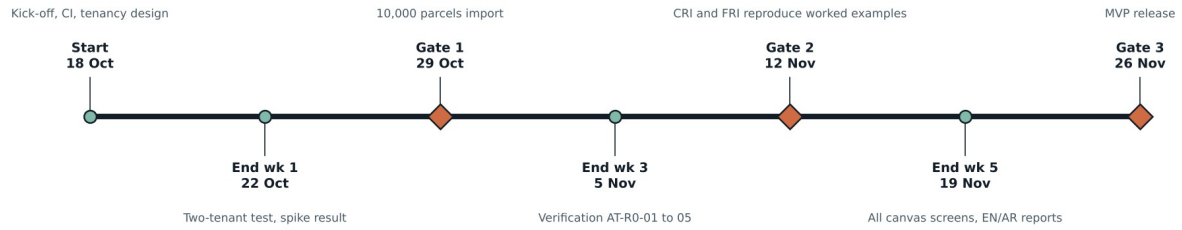
`{tenantId, farmId}`, `{tenantId, parcelId, season}`, `2dsphere` on boundaries, cursor pagination everywhere, portfolio numbers read from aggregates, scene statistics cached per parcel and scene. Shard key `{tenantId, _id}` is reserved, not used yet. A throughput target is set after the raster spike (task W1-5).

Fast-track build order

Six weeks, assuming one lead developer working with Claude Code full time; if the team is bigger, run the parallel tracks below. Gates are only at the end of weeks 2, 4 and 6. Inside a week, work in the order listed and ship each item behind its test.

Timeline

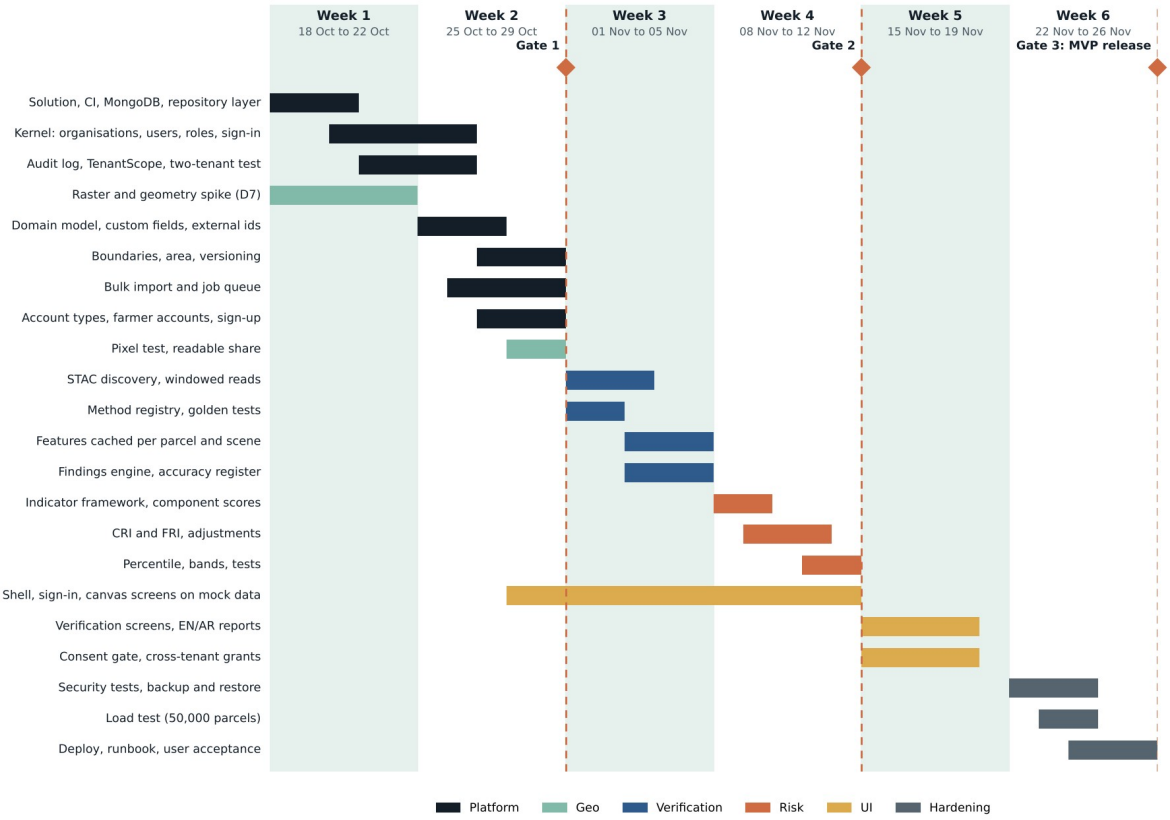
Start Sunday 18 October 2026 (work week Sunday to Thursday, an assumption); release Thursday 26 November 2026.



Timeline: start 18 Oct, Gate 1 on 29 Oct, Gate 2 on 12 Nov, Gate 3 on 26 Nov

Gantt chart

Bars are working days. Platform, Geo, Verification, Risk, UI and Hardening are separate tracks, so the UI bars can start in week 2 against mock data.



Gantt chart of the six-week MVP plan by track, with three gates

Weekly detail

Week	Build	Done when
1. Foundation 18 to 22 Oct	W1-1 solution, CI, lock files, pre-commit secret scan. W1-2 MongoDB single-node replica set with auth, repository layer, migration runner, append-only role. W1-3 Kernel: organisations, users, memberships,	CI green. Two-tenant test passes. Spike matches QGIS or the Python sidecar is chosen

Week	Build	Done when
	roles, permissions, invitation sign-in (email, password, TOTP). W1-4 audit log and `TenantScope` with the two-tenant test. W1-5 raster and geometry spike (D7): one parcel window from a Sentinel-2 COG via GDAL, area and pixel count, compared with QGIS	
2. Domain and import 25 to 29 Oct	W2-1 Portfolio, Farm, Parcel, ParcelSeason, custom fields, external ids. W2-2 boundary validation, versioning, area in EPSG:32636. W2-3 bulk import job: staged, validated, idempotent by external id, error report. W2-4 job queue with per-tenant fairness. W2-5 pixel test and readable share by size class. W2-6 AccountType presets, farmer accounts (direct, or managed by a company or authority), sign-up and approval by account type, farmer consent grants with Withdraw	**Gate 1 (29 Oct):** 10,000 synthetic parcels import in a background job, errors listed, re-import creates no duplicates
3. Imagery and Verification 1 to 5 Nov	W3-1 STAC discovery, windowed reads of B2, B3, B4, B8, SCL, masks, provenance. W3-2 NDVI and other features, cached per parcel and scene. W3-3 method registry with freeze and golden-parcel test. W3-4 findings engine v0: area gap with noise, crop consistency, two-pass rule, `NOT_ASSESSABLE`. W3-5 accuracy register, status and `accuracy_ref` on every finding	AT-R0-01 to AT-R0-05 pass
4. Risk module 8 to 12 Nov	See the Risk module section	**Gate 2 (12 Nov):** CRI and FRI computed for the synthetic book and one real parcel-season; hand-check matches the methodology worked examples
5. UI and reports 15 to 19 Nov	W5-1 shell, sign-in, access request, language switch. W5-2 Portfolio, Farm, Farmers, Parcel risk, Methodology, Import, Farmer home, Insurance event evidence and Account types screens from the design canvas. W5-3 Verification screens: analysis, imagery, accuracy. W5-4 reports in English and Arabic (PDF, CSV), frozen with SHA-256 manifest. W5-5 consent gate and cross-tenant grants	Every canvas board has a working screen or is listed as deferred
6. Hardening and release 22 to 26 Nov	W6-1 security tests: cross-tenant, consent bypass, leakage, rate limits. W6-2 backup and restore test. W6-3 load test with 50,000 parcels. W6-4 deploy to the chosen host, monitoring, runbook. W6-5 user acceptance with the first customer	**Gate 3 (26 Nov), MVP release:** definition of done below

Risk module (CRI and FRI), week 4

Implement exactly what the CRI/FRI methodology specification (FRI-2026-01) says. Do not tune weights or cut points. v1 said not to build this; v2 reverses that for the MVP, under the rules below.

Task	Detail	Done when
R-1 Indicator framework	Indicator definition, immutable result (method version, input hash, value, components, missing components, maturity), recompute creates a new result	Recomputing with the same inputs gives the same hash and value
R-2 Component scores	Seven components with weights 25, 20, 15, 15, 10, 8, 7. A component without data is stored as not established with its weight, never as zero	Parcel ZR-KS-03 reproduces CRI 50.3
R-3 CRI	Weighted mean renormalised over established weight; validity floor excludes parcels from FRI	Unit tests for 7, 5 and 3 established components
R-4 FRI	Area-weighted mean of parcel CRIs plus five adjustments: crop concentration, spatial concentration, irrigation diversity, record length, worst-parcel exposure	Delta Agro Estate reproduces AWCRI 38.7, adjustments +1.7, FRI 40.4
R-5 Percentile and band	Percentile in a reference cohort; bands A to E at 0, 20, 40, 60, 80	Band is a pure function of percentile, tested at every cut point
R-6 API and screens	Portfolio, Farm, Parcel risk, Methodology, with the maturity badge on every number	Matches the canvas boards
R-7 Reports	Farm and parcel risk pages in English and Arabic	Frozen and reproducible like Verification reports

Rules for this module

- Every CRI or FRI value, screen, export and report shows "Specified · not calibrated" and the method version. Calibration is scheduled for 2026/27.
- Never call it a credit score, probability of default, approval or recommendation. No ranking page, and band colours are never used as verdict colours.
- Several component inputs (soil, climate, tenure, yield history) need data sources the team has not named. Until a source is connected, that component is not established and the weight is renormalised. This is allowed by the method and keeps the MVP unblocked.
- Known specification gap: the worked example gives FRI 40.4 as band B, but 40.4 sits between the published cut points (Gap A-2). Implement bands on percentile as specified in R-5 and record the decision in an ADR.

Parallel tracks, definition of done and open decisions

Run in parallel to finish sooner

Track	Work	Owner
A. Platform	Weeks 1, 2 and 6: kernel, tenancy, import, jobs, hardening	Lead developer
B. Geo and Verification	Weeks 1, 3: spike, imagery, features, findings	Second developer or the same one after week 2
C. Risk	Week 4, can start in week 3 against synthetic features	Developer with the methodology open
D. UI and reports	Week 5, can start in week 2 against mock data from the canvas	Frontend developer
E. Inputs and legal	Data sources for the risk components, licence check for Sentinel use (IMG-07), Arabic glossary review, consent wording with counsel	Founders

Definition of done for the MVP

- Two organisations cannot read each other's data; the isolation test runs in CI.
- 50,000 synthetic parcels import, process and show in the portfolio within the agreed time (set after the spike).
- Every Verification finding has a status and `accuracy_ref`; "below resolution" is never shown as "no crop".
- Every CRI and FRI shows the maturity badge and method version; worked examples reproduce.
- Reports regenerate from their recorded snapshot and match their SHA-256, in English and Arabic.
- Consent, grants and audit are tested, including refusals.
- Backup and restore tested; runbook written.

Open decisions and risks

- **Team size and dates.** Six weeks assumes one full-time developer with Claude Code and the inputs arriving on time. The start date of 18 October is an assumption.
- **Risk inputs.** Which sources feed soil, climate, tenure, yield history and management quality. Without them the index will show few established components.
- **Throughput target.** Set after the week-1 raster spike; a failed spike adds a Python sidecar for window reads.
- **Hosting.** Local MongoDB is fine for development; the host and its licence check are needed before week 6.
- **Counsel.** Consent chain and wording under Law 151/2020, and review of the wording of the risk labels.
- **Arabic.** All Arabic text in the canvas needs a native reviewer.
- **Spec gap A-2.** Band for values between cut points (see Risk module).